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Tree Species Identification using Deep Learning

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Abstract

Accurate identification of individual tree species is essential for forest management, biodiversity assessment, and ecosystem monitoring. Although great progress has been achieved in object detection and classification tasks of tree crowns with deep learning models in recent years, most existing recognition models still use a single-modal input with low-dimensional representations to some extent.

In this work, we propose the TreeAI-Swiss dataset with a multi-modal analysis at the individual tree level for Switzerland. The proposed dataset combines high-resolution aerial images, topographical maps, multi-temporal Sentinel-1 and Sentinel-2 images, Google Satellite Embeddings, and bioclimatic variables. The sparse and reliable instance labels for the trees are obtained through a semi-automatic process based on field measurements and crown delineation with SAM3. An ILP-based splitting algorithm is used for handling the spatial dependence of the data.

Building on this dataset, we develop a multi-modal fusion network using a frozen foundation model AnySat with encoders. Detection and classification tasks are conducted jointly while adapting weights for modalities based on a novel learnable gate. Experiments on various modal combinations and temporal scales show that multi-modal fusion improves the performance for detection as well as classification compared to the baselines on the RGB channel alone. Per-species analysis reveals significant performance on dead tree detection, proving the applicability of the designed approach for forest health monitoring.

Overall, this work presents a comprehensive dataset, modeling framework, and experimental analysis for multi-modal, instance-level forest monitoring, and provides a flexible foundation for future research on adaptive fusion strategies and large scale ecological assessment. Our code and models are available at `https://github.com/yixin-zhou/Tree_Species_Identification`

Chapter 1

Introduction

Forests are critical components of the Earth system. Climate regulation, biodiversity conservation, or ecosystem services all have the forests at their core. This means that accurate, up-to-date information about the existence of these ecosystems is essential to their sustainable management (Ouaknine et al., 2024). The emergence of these aerial platforms with diversity in sources of remote sensing data highlights the need for effective approaches to transfer these sources of information into meaningful variables that describe the ecosystems of interest (Hemmerling et al., 2021).

Forest monitoring methods developed so far are mostly based on single or dual type remote sensing observations only, for instance, high-resolution RGB images (Schiefer et al., 2020) or optical satellite observations (Hemmerling et al., 2021). Their performance is encouraging for tree detection as well as tree species mapping but is limited by the spatial resolution as well as the available observation content provided by individual sensors. A single pixel or observation often contains aggregated information from multiple trees belonging to different species and canopy layers. As a result, tree-specific structural details are partially degraded at the individual-tree scale due to spatial averaging. This limitation poses challenges for scalable representations aimed at extracting fine-grained tree-level information from multi-source remote sensing observations.

In this respect, individual tree crown detection and delineation constitute a fundamental component of forest remote sensing. Individual tree crowns serve as a basic reference unit for quantifying forest structure, productivity, and ecosystem functions at the per-tree level (Freudenberg et al., 2022).

Despite their precision, field-based measurements are highly labor-intensive and therefore impractical for large-scale and repeatedly updated forest monitoring. This limitation has driven increasing interest in remote sensing–based approaches for forest observation (Safonova et al., 2021a). With the growing availability of very-high-resolution remote sensing imagery, individual tree crown delineation has become a prerequisite for translating pixel-level measurements into meaningful tree-level representations (Troles et al., 2024).

Recent breakthroughs in the field of deep learning have led to significant improvements in the accuracy of tree crown detection and segmentation compared to traditional image processing approaches (Galuszynski et al., 2022). However, achieving reliable and adaptable crown detection across diverse forest types and sensor conditions remains a challenging task. This difficulty largely arises because many existing models depend on a limited data representation or a single sensing modality (Schiefer et al., 2020; Hemmerling et al., 2021).

More specifically, reliance on individual modalities makes it difficult to capture complementary forest canopy information at the individual tree crown level through joint processing in deep learning models (Freudenberg et al., 2022). As a result, such approaches struggle to fully exploit the heterogeneous structural and spectral characteristics of forest canopies. These limitations clearly indicate the need for appropriate data formats and modeling paradigms to enable efficient progress in tree crown recognition, particularly in complex and dense forest settings. Accordingly, there is a growing necessity to develop multi-modal remote sensing datasets with an explicit focus on individual tree crown recognition. Such datasets are essential for preserving high-resolution characteristics of individual trees that cannot be adequately captured using pixel-level or single-modal sensing alone. At the same time, the effective use of these high-dimensional data sources requires learning models capable of integrating multiple modalities in a coherent manner. This motivates the development of multi-modal fusion learning models specifically tailored for individual tree crown analysis.

Chapter 2

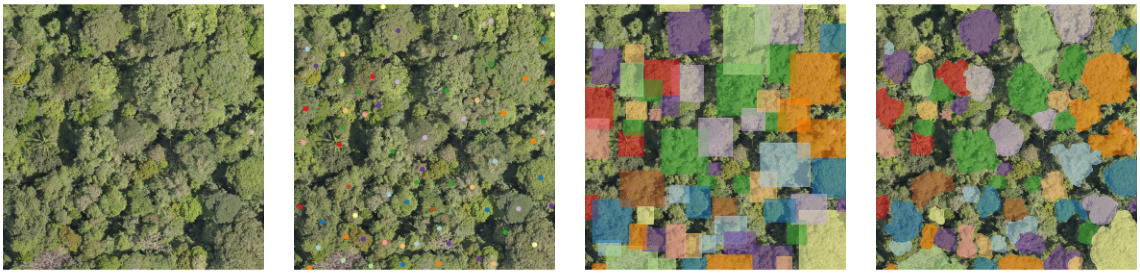
Related Works

2.1 Aerial-based Datasets

Aerial datasets for forest-related tasks adopt a variety of annotation paradigms that differ in both supervision level and geometric expressiveness, as illustrated in Figure 2.1. Patch-level annotations assign a single label to an image region and are primarily used for forest type mapping or tree species classification, but they do not provide information at the individual-tree level. Point- and bounding-box annotations introduce coarse object-level supervision and are commonly employed for individual tree detection, yet they still offer only limited geometric detail of tree crowns.

In contrast, polygon-based annotations explicitly delineate canopy boundaries. Depending on whether contiguous canopy regions or individual tree crowns are annotated, polygon annotations can support either semantic segmentation or instance segmentation. Overall, these annotation paradigms reflect a clear trade-off between annotation cost and representational accuracy. In particular, instance-level polygon annotations provide the most precise description of individual tree crowns, but require substantially higher annotation effort.

Building upon the annotation paradigms discussed above, a wide range of aerial image-based forest datasets have been proposed to support tree-related monitoring tasks. TreeSatAI (Ahlswede et al., 2023) is a large-scale aerial forest dataset that combines high-resolution aerial imagery with Sentinel-1/2 time series to support patch-level tree species classification across diverse forest types. SELVABOX dataset (Baudchon et al., 2025), collected across tropical regions in Panama, Brazil, and Ecuador, provides bounding-box annotations for individual tree detection in rainforest environments. But it lacks fine-grained crown delineations required for



(a) Patch-level **(b)** Point-based **(c)** Bounding-box **(d)** Polygon-based

Figure 2.1: Comparison of different annotation methods. (a) Labels an entire image region with a single class. (b) Marks representative GPS points for each class. (c) Encloses objects with rectangular boxes. (d) Outlines object boundaries. In terms of task type, (a) belongs to *Tree Species Classification*, (b) and (c) to *Individual Tree Detection*. For (d), if the polygons delineate contiguous forest stands or canopy patches, the task is categorized as *Semantic Segmentation*; however, if the polygons precisely trace the crown boundary of each individual tree, it is regarded as *Instance Segmentation*.

instance-level analysis. Quebec Trees Dataset (Cloutier et al., 2024) is specifically designed to evaluate tree detection and instance segmentation methods for diverse urban and peri-urban forest structures but only based on RGB images. Global-GeoTree (Mu et al., 2025) is a global, multimodal tree species classification dataset with over 6 million samples, integrating Sentinel-1/2 time series, environmental variables, and hierarchical taxonomic labels. OAM-TCD dataset (Veitch-Michaelis et al., 2024) is an open-source dataset derived from UAV imagery for urban tree crown detection, featuring instance-level annotations but lacking species-level semantic information required for fine-grained ecological analysis.

A comprehensive comparison of existing aerial image-based forest datasets, including their geographic coverage, spatial resolution, annotation granularity, and target tasks, is provided in Appendix A.1. Despite the increasing availability of forest datasets, benchmarks that simultaneously provide multi-modal data and high-quality instance-level annotations of individual tree crowns remain limited.

2.2 Tree Crown Detection and Segmentation

Tree crown detection and segmentation have been widely explored using deep learning techniques applied to high-resolution aerial and UAV imagery. Early approaches typically formulate the problem either as an object detection task or as a semantic segmentation challenge, in which individual tree crowns are identified using convolutional neural network architectures. For example, Beloiu et al. (Beloiu et al., 2023) demonstrate the use of Faster R-CNN on aerial RGB imagery to effectively locate and classify individual trees in heterogeneous temperate forests. In parallel, Ma et al. (Ma et al., 2025) introduce an optimized algorithm for tree crown segmentation in UAV images, designed for neural network accelerators to balance segmentation accuracy with computational efficiency.

Beyond single-image methods, recent research has focused on incorporating more sophisticated spatial and temporal features to improve crown delineation accuracy. Firoze et al. (Firoze et al., 2023) propose a temporal contour graph to better represent the dynamics of crown appearance in image sequences, making instance segmentation more robust to crown occlusions and overlaps. Ramesh et al. (Ramesh et al., 2024) investigate tree semantic segmentation using airborne image time series and verify that integrating phenological information improves segmentation accuracy compared to single-date imagery. Their study further shows that learning with taxonomic supervision at multiple levels of granularity, including species, genus, and higher taxonomic groups, can enhance segmentation performance. Speckenwirth et al. (Speckenwirth et al., 2024) introduce a tool called *TreeSeg* for fully automatic tree crown segmentation from high-resolution multispectral UAV data. Voulgaris (Voulgaris, 2025) propose *PerceptiveNet*, a deep neural network model that combines traditional image processing knowledge with modern deep learning architectures.

2.3 Remote Sensing Foundation Models

In recent years, there has been tremendous progress in remote sensing foundation models that aim to derive general-purpose knowledge from a massive scale of Earth observation data and apply this knowledge to a wide range of downstream applications. Early benchmarks like BigEarthNet (Sumbul et al., 2019) proved that large-scale pretraining is applicable to remote sensing image understanding in a general-purpose manner mainly targeting single-modal optical imagery and associated per-scene classification tasks. However, continuing this trend, recent work in remote sensing foundation models has also shifted to a growing emphasis on multimodal learning to effectively utilize complementary cues across multiple sensors. For example, models like OmniSat (Astruc et al., 2024) and AnySat (Astruc et al., 2025) introduce self-supervised architectures that incorporate modality-agnostic inputs with multiple sensors to support a unified backbone architecture to generalize across modalities and spatial resolutions. Meanwhile, SkySense (Guo et al., 2024) also explored large-scale pretraining in a multi-modal setting for genuine universal remote sensing interpretation. GeoPix (Ou et al., 2025) and

Foundation models offer an attractive avenue for dealing with the inherent heterogeneity of forest ecosystems, where structural, spectral, and temporal features are combined in a complex manner that jointly shapes forest structure and dynamics. FoMo Bountos et al. (2025) represents an interesting attempt at building a bridge between foundation models and domain-specific tasks in the context of forests, merging multi-modal and multi-scale information from remote sensing data.

However, despite their strong representational capabilities, most existing remote sensing foundation models are primarily trained and evaluated on scene-level or other coarse-grained tasks. As a result, their applicability to fine-grained, instance-level forest analysis remains insufficiently explored. This limitation is particularly evident for tasks at the level of individual tree crown identification. Consequently, it remains an open question whether the proposed multi-modal representations can be effectively transferred to tree-level forest analysis, especially in settings where high-quality instance-level annotations are available.

Chapter 3

Data and Dataset Preprocessing

The current chapter explains the creation and pre-processing of the TreeAI-Swiss dataset, which is a multi-sensory dataset for analyzing individual trees using a range of remote sensing techniques. Because the use of conventional RGB imagery alone can result in a degree of ambiguity for tree species classification, especially in complex forest structures where various tree species can have a similar appearance (Schiefer et al., 2020), additional information from various sources is necessary to provide complementary information for the purpose of differentiation (Mu et al., 2025). This need for complementary information motivated the construction of a multi-sensory dataset that explicitly focuses on the spatial context surrounding individual tree crowns. By targeting tree-level regions, the dataset enables the development of models that can jointly exploit different sources, as illustrated in Figure 3.1.

Figure 3.3 shows example instances of the various data modalities in TreeAI-Swiss dataset. A summary of the band composition and the spatial resolution of each dataset is provided in Table 3.1.

3.1 Data

3.1.1 High-Resolution Aerial Orthophoto Mosaic

The most important type of data in the TreeAI-Swiss dataset is the high-resolution (HR) aerial orthophoto image, since it directly records the visual and spatial information of the tree crowns from a overhead view. The information from this type of data is highly relevant for distinguishing tree species based on visual patterns of the crowns of the trees. The aerial orthophoto images of the trees in the TreeAI-

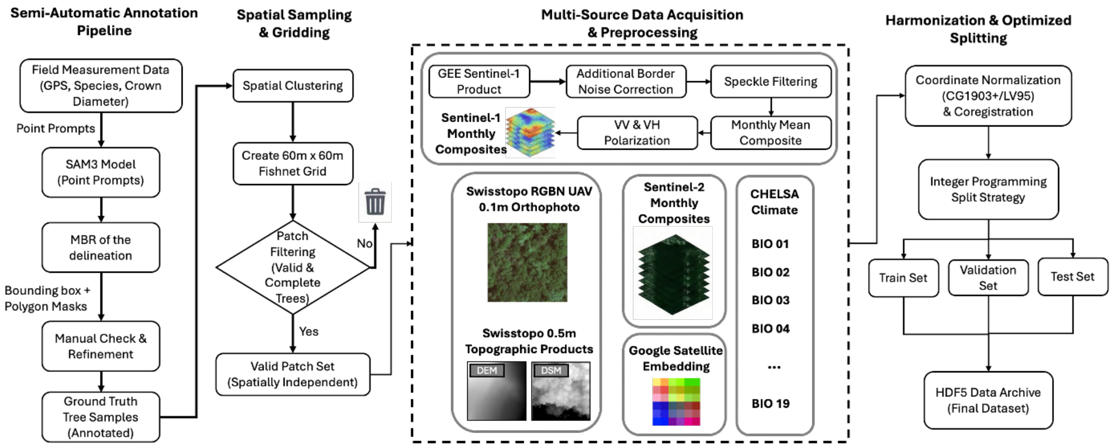


Figure 3.1: Overview of the TreeAI-Swiss data construction and preprocessing pipeline. Multi-modal remote sensing inputs are collected, spatially aligned, filtered, and transformed into a unified dataset for individual tree crown analysis.

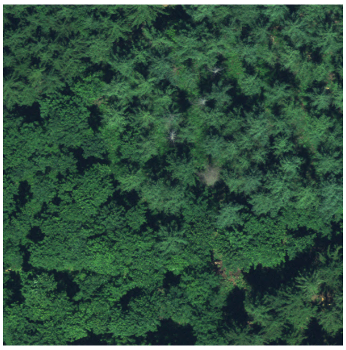


Figure 3.2: Example of high-resolution aerial orthophoto imagery illustrating canopy-level structural differences. Tree crowns in the upper-left and lower-right regions exhibit clearly distinct canopy structures in crown shape and texture.

Swiss dataset were sourced from the SWISSIMAGE dataset of Swisstopo (Swisstopo, 2024b).

The orthophoto mosaic SWISSIMAGE 10 cm is a composition of new digital color aerial photographs over the whole of Switzerland with a ground resolution of 10 cm in the plain areas and main alpine valleys and 25 cm over the Alps.

3.1.2 Topographic Data

Topographic data provide important contextual information for both tree species distribution and individual tree crown delineation. In TreeAI-Swiss, digital elevation models (DEM) capture elevation-related environmental conditions that influence tree growth, as many tree species exhibit characteristic distributions along altitude gradients (Körner, 2007). DEM information therefore offers an ecological constraint that supports tree species discrimination beyond RGB appearance.

Digital surface models (DSM), in contrast, encode height-related structural information of the canopy. In dense forests, overlapping crowns often appear visually homogeneous in RGB imagery, making individual tree boundaries difficult to distinguish. High-resolution DSMs provide complementary vertical cues that help separate adjacent crowns, and recent studies have shown that integrating RGB and DSM information improves tree crown segmentation, including SAM-based approaches (Teng et al., 2025).

Digital elevation information and canopy surface height information are obtained from the swissALTI3D dataset (Swisstopo, 2024a) and swissSURFACE3D Raster dataset (Swisstopo, 2024c), both provided by Swisstopo.

3.1.3 Sentinel Remote Sensing Data

Satellite-based remote sensing provides complementary data for high-resolution aerial images by observing spectral, temporal, and structural patterns of tree canopies over larger regions. In the TreeAI-Swiss dataset, data from Sentinel-2 and Sentinel-1 satellites is combined to provide multispectral and radar data for tree-level evaluations.

Sentinel-2 optical data contains a range of broad bands such as visible, near-infrared, red-edge, and short-wave infrared bands. The red-edge and short-wave infrared bands of Sentinel-2 data were proven to be effective for tree species classification, with benefits to tree species identification through Sentinel-2 data (Immitzer et al., 2016). In addition, the revisit cycle of Sentinel-2 data has proven to be efficient for the incorporation of seasonal differences in vegetation phenology to improve species classification accuracy compared to the use of single-date data (Axelsson et al., 2021).

The data from the Sentinel-1 synthetic aperture radar (SAR) missions offers a set of information regarding structure and moisture, largely independent of cloud cover and illumination conditions. Multi-temporal data from the Sentinel-1 mission has been successfully used for distinguishing forest types and tree species based on differences in structure and radar backscatter behavior over time (Udali et al., 2021). A combination of data from both Sentinel-1 and Sentinel-2 has also been shown to enhance the mapping of large areas of tree species (Blickensdörfer et al., 2024).

Although the Sentinel datasets have proved useful for the regional-scale observation of the forest cover, they suffer from moderate spatial resolution, thereby

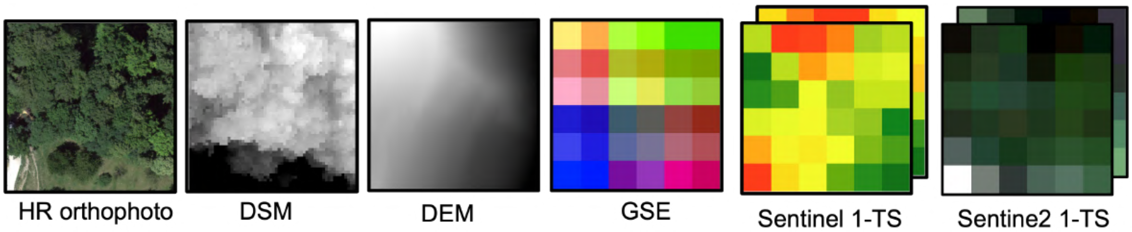


Figure 3.3: Example of multi-modal data representations used in the TreeAI-Swiss dataset. From left to right: HR aerial RGB imagery, DSM, DEM, Sentinel-2 time series data, Sentinel-1 time series data, and GSE representations.

hindering their ability to spatially resolve the crowns of the individual trees. In the TreeAI-Swiss system, the Sentinel-1 and Sentinel-2 data are merged with the high-resolution aerial images.

3.1.4 Bioclimatic Variables

Climatic conditions play a fundamental role in shaping tree species distributions by defining species-specific ecological conditions. Temperature and precipitation regimes, as well as their seasonal variability and climatic extremes, are widely recognized as key environmental drivers of species occurrence and habitat suitability (Guisan and Zimmermann, 2000).

HR climate predictors have been shown to substantially improve species distribution modelling performance (Karger et al., 2017). CHELSA-BIOCLIM (Brun et al., 2022) derives a set of bioclimatic variables designed to summarize annual means, seasonality, and climatic extremes that are ecologically meaningful for species distribution. Accordingly, TreeAI-Swiss incorporates CHELSA BIOCLIM variables as auxiliary environmental information to support tree species classification.

3.1.5 Google Satellite Embedding Dataset

The Google Satellite Embedding (GSE) dataset is derived from the AlphaEarth Foundations model (Brown et al., 2025), which learns a general geospatial embedding by assimilating spatial, temporal, and measurement contexts across multiple Earth observation.

We use the released global, annual, analysis-ready embedding layers as auxiliary features to provide high-level contextual information that complements pixel-level spectral, structural, and environmental variables in TreeAI-Swiss.

Table 3.1: Overview of data modalities used in the TreeAI-Swiss dataset, including band composition and spatial resolution.

Data Modality	Number of Bands	Spatial Resolution
Aerial orthophoto	4 (RGB + NIR)	0.1 m
Digital Elevation Model (DEM)	1 (elevation)	0.5 m
Digital Surface Model (DSM)	1 (surface height)	0.5 m
Sentinel-2 optical imagery	10 (multispectral)	10 m
Sentinel-1 SAR imagery	2 (VV, VH)	10 m
Google Satellite Embeddings (GSE)	64 (embedding features)	10 m

3.2 Dataset Preprocessing

The pipeline covers annotation generation, spatial sampling, multi-source data preprocessing, and data harmonization to ensure spatial consistency and reliable alignment across data modalities described in section 3.1. Annotation generation is based on a combination of field-observed reference points. A semi-automatic labeling workflow, built upon SAM-based models, enables efficient individual tree crown annotation. Multi-source data preprocessing focuses on transforming raw remote sensing products into analysis-ready inputs, including radiometric and geometric preprocessing of Sentinel-1 SAR imagery and monthly compositing of cloud-free Sentinel-2 optical time series. Data harmonization further aligns all modalities by reprojecting them into a unified coordinate reference system and performing coregistration to ensure pixel-level correspondence. In addition, an integer linear programming based dataset splitting strategy is adopted to address the highly imbalanced and long-tailed species distribution of the dataset.

3.2.1 Semi-Automatic Annotation Pipeline

The semi-automatic annotation pipeline of the TreeAI-Swiss dataset is built upon high-quality field measurement data collected by professional forestry experts. Tree species information is obtained through long-term and repeated field surveys, during which each tree is carefully identified and recorded together with its geographic location and crown-related attributes. Given the expertise of the observers and the stability of the observation protocol, the resulting species labels are considered highly reliable. Therefore, these in-situ observations serve as the ground truth labels of the dataset.

Segment Anything Model(SAM) proposes a promptable segmentation frame-

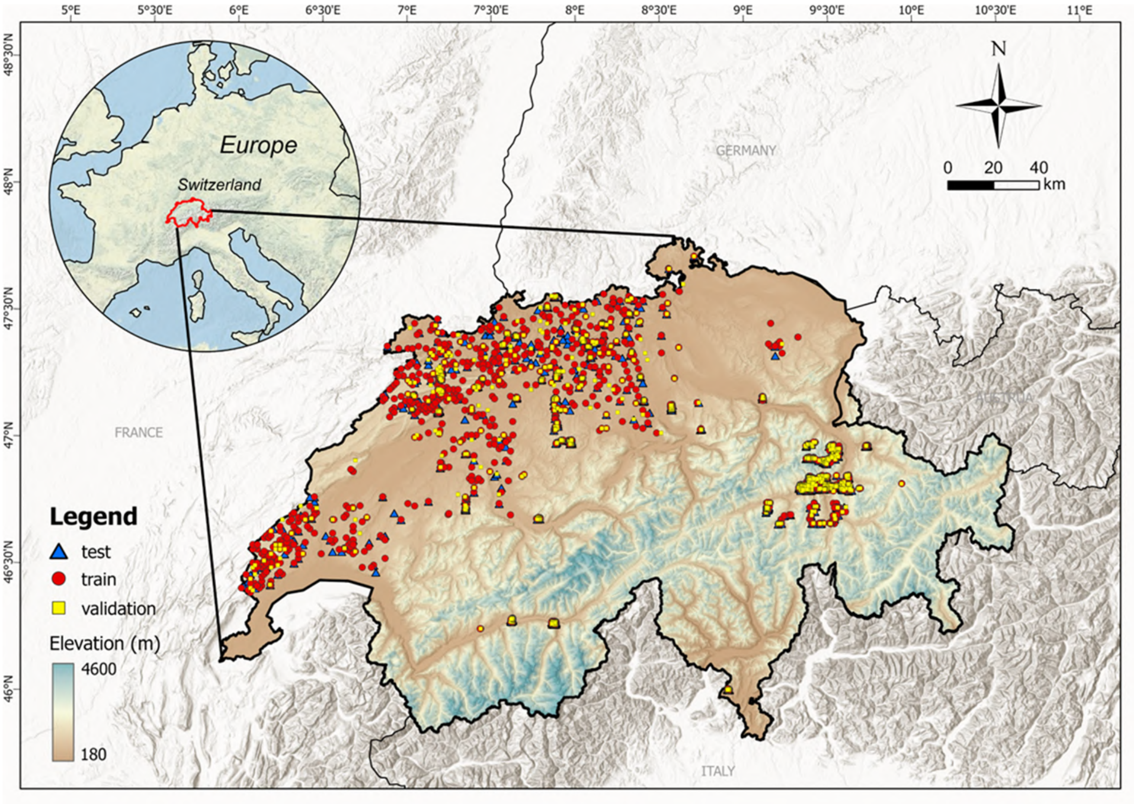


Figure 3.4: Overview of the study area and spatial distribution of samples in Switzerland. Colored markers indicate the locations of training, validation, and test samples.

work that can generate object masks conditioned on simple user inputs such as points, bounding boxes, or masks. By training on the large-scale dataset, SAM demonstrates strong generalization ability and establishes a foundation model paradigm for image segmentation (Kirillov et al., 2023).

SAM3 (Carion et al., 2025), as the latest model of SAM family, further introduces concept-level and textual prompting, allowing segmentation to be guided by high-level semantic descriptions (e.g. "tree crowns").

Given that the field measurement dataset contains more than 20,000 individual tree records, manually delineating tree crown polygons for each sample is impractical. To address this challenge, we adopt a semi-automatic annotation strategy that leverages the strong zero-shot segmentation capability of SAM3 to efficiently generate high-quality crown delineations.

First, the field-measured GPS points associated with tree species labels are first overlaid onto the HR aerial orthophoto imagery (Section 3.1.1). Each GPS point is then used as an individual point prompt to guide SAM3 in segmenting the corresponding tree crown, complemented by a textual prompt ("tree crowns") to provide

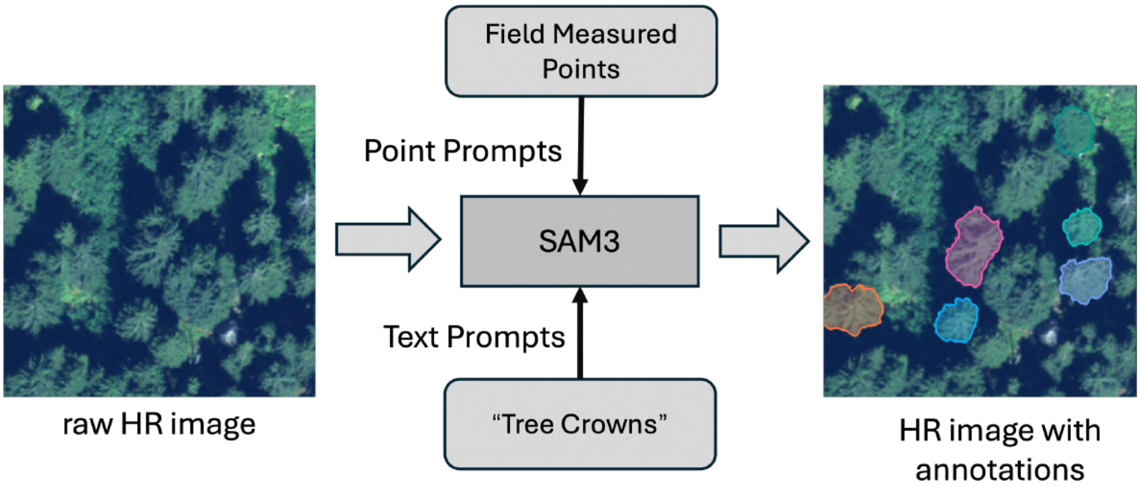


Figure 3.5: Semi-automatic tree crown annotation using SAM3 guided by field-measured point prompts and a textual prompt (*"tree crowns"*).

semantic context. Each GPS point is treated as a distinct object, ensuring instance-level crown delineation.

To ensure annotation quality, a manual verification step is incorporated into the pipeline. In rare cases where SAM3 produces visibly incorrect segmentations, particularly in dense or highly overlapping canopy regions, human intervention is applied to correct or refine the generated masks. In practice, such corrections occur infrequently, because of the robustness zero-shot performance of SAM3 in forest scenes.

Finally, the minimum bounding rectangles (MBR) of the validated segmentation masks are extracted as tree crown bounding boxes. This allows the dataset to support not only instance-level crown segmentation but also tree species detection tasks within a unified annotation framework.

3.2.2 Spatial Sampling and Gridding

As illustrated in Fig. 3.4, sample points in the TreeAI-Swiss dataset are distributed across the entire territory of Switzerland. To transform these point-based observations and their associated crown delineations into a representation suitable for deep learning models, the samples must be organized into spatially bounded image patches of fixed size. In forest remote sensing, a patch size of $60\,\mathrm{m} \times 60\,\mathrm{m}$ is commonly adopted across multi-modal datasets (such as Mu et al. (2025); Ahlswede et al. (2023)). Accordingly, TreeAI-Swiss follows this convention and adopts a uniform patch size of $60\,\mathrm{m} \times 60\,\mathrm{m}$ for all downstream tasks.

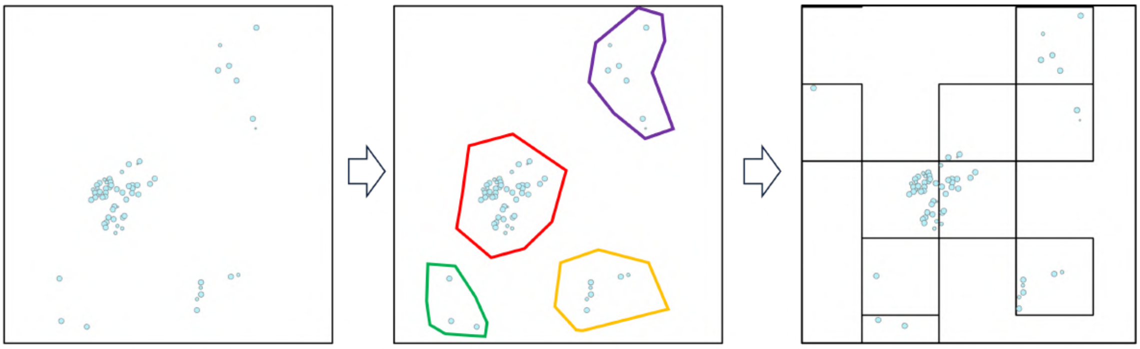


Figure 3.6: Sample points are first grouped using K-means clustering. For each cluster, a 60 m × 60 m fishnet grid is generated to cover all contained samples, and grid cells without any sample points are subsequently discarded.

A straightforward solution would be to generate a nationwide 60 m × 60 m fishnet covering the entire territory of Switzerland. However, given the large spatial extent of the country, such an approach would result in a massive number of grid cells, most of which do not contain any samples, leading to substantial and unnecessary memory consumption.

To address this issue, we design a memory-efficient spatial sampling and gridding strategy tailored to the actual distribution of sample points. First, all field-measured sample points are spatially clustered using a K-means algorithm, which is performed based on the planar coordinates of the samples. This step groups geographically close samples into a set of spatially coherent clusters.

For each cluster, a local fishnet grid with a fixed cell size of 60 m × 60 m is then generated. Compared to a nationwide gridding scheme, this localized strategy significantly reduces memory usage by avoiding the creation of a large number of empty grid cells. In addition, it produces grids that better align with the underlying sample distribution. This means that multiple nearby samples may likely to fall within the same grid cell.

This design also facilitates dataset splitting, as grid cells are non-overlapping and spatially independent. Finally, any grid cells that do not contain sample points are discarded, yielding a compact and efficient set of spatial patches for downstream processing. Fig. 3.6 and algorithm 1 provides a visual overview of the key steps of the proposed method.

Algorithm 1: Spatial sampling and gridding for TreeAI-Swiss.

Input: Sample points with coordinates (x, y) and labels, grid size $60\,\mathrm{m} \times 60\,\mathrm{m}$.
Output: A set of valid spatial patches for model training.
Cluster all sample points using K-means based on their spatial coordinates
foreach *cluster* **do**
    Generate a local $60\,\mathrm{m} \times 60\,\mathrm{m}$ fishnet covering the cluster extent
    Remove grid cells that contain no sample points
return *All remaining grid cells as dataset patches*

3.2.3 Multi-Source Data Preprocessing

Sentinel-2 Time Series

Most Sentinel-2 downstream land applications implicitly assume that cloud and cloud-shadow contaminated pixels have been identified and removed prior to downstream analysis (Main-Knorn et al., 2017).

For Sentinel-2 imagery, we employ the harmonized Sentinel-2 surface reflectance product [1] available on Google Earth Engine (GEE). Ten optical bands are used in this study and they are all resampled to a unified ground sampling distance (GSD) of 10 m.

Tree species exhibit pronounced seasonal phenological differences. To capture such seasonal dynamics, our objective is to construct cloud-free monthly Sentinel-2 composites over a full annual cycle. Cloud masking is performed using the Scene Classification Layer (SCL). Only pixels corresponding to vegetation, bare soil, and water surfaces are retained, while all other classes are excluded. For each month, cloud-free composites are generated using a median compositing strategy.

Sentinel-1 Time Series

For Sentinel-1 preprocessing, we used the Sentinel-1 GRD collection [2] available in GEE and applied an analysis-ready data preparation workflow stritcly following the method proposed by Mullissa et al. (2021). Sentinel-1 GRD imagery is affected by border noise artifacts and inherent speckle noise, which can negatively impact downstream analysis if not properly addressed. Therefore, we applied additional border noise correction, multi-temporal speckle filtering using a Gamma MAP filter,

[1] `https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED`
[2] `https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD`

Table 3.2: Sentinel-1 SAR preprocessing configuration used in this study based on the framework proposed by Mullissa et al. (2021).

Parameter	**Setting**
Data source	Sentinel-1 GRD (GEE: COPERNICUS_S1_GRD)
Acquisition mode	Interferometric Wide Swath (IW)
Orbit direction	Descending
Polarization	VV and VH
Data format	Linear scale
Additional border noise correction	Enabled
Speckle filtering framework	Multi-temporal
Speckle filter type	Gamma MAP
Speckle filter kernel size	9×9
Number of images for filtering	10
Radiometric terrain normalization	Enabled
Terrain flattening model	Volume scattering
Layover and shadow masking	Enabled (no additional buffer)
Temporal compositing	Monthly mean aggregation

and radiometric terrain normalization with a volume scattering model. The resulting backscatter images were aggregated using monthly mean compositing for subsequent analysis.

Other

For the remaining data sources, the acquisition time of each field-measured GPS point is used as the temporal reference, and data from all modalities are selected to be as temporally close as possible. Using the patch extent defined in Section 3.2.2, DEM, DSM, and GSE are clipped to the corresponding $60\,\text{m} \times 60\,\text{m}$ patch area. For CHELSA BIOCLIM, the center of each patch is taken as the sampling location, from which the 19 bioclimatic variables are extracted. With the patch area of 60 m $\times$ 60 m, it can be presumed that the bioclimatic values at the sampling location represent the patch bioclimatic conditions in their entirety.

3.2.4 Harmonization

The TreeAI-Swiss dataset is a combination of the data sources that come from the different spatial systems and geometries, which require a process of harmonization.

Swisstopo offers high resolution data products such as DEM, DSM, and HR orthophoto mosaic, in the CH1903+/LV95 map coordinate system. These data are corrected for terrain using ground control points, so they can be thought of as well coregistered. This means pixels in corresponding images in geographic space are in alignment.

In contrast to this, the medium-resolution data sources derived from Google Earth Engine (GEE), namely Sentinel-1, Sentinel-2, and GSE, provide data in the WGS-84 coordinate reference system. The derived data sources are reprojected to CH1903+/LV95 to provide a uniform reference framework for all modalities. Because of the differences in satellite observation angles and observation heights, there is no guarantee of accurate pixel registration performed on pixel-level correspondences between the satellite imagery of 10 m resolution and high–resolution aerial imagery sources. However, based on the coarse resolution of satellite imagery with a fixed size of the analyzed image patch of 60 m $\times$ 60 m, the spatial accuracy tolerance value of few meters is acceptable. This means that we did not do the accurate registration of satellite imagery sources to Swisstopo data sources. To further enable efficient storage and loading in a training environment in turn, all harmonized multi-modal data sources are stored in an HDF5 file.

3.2.5 Dataset Splitting

Fig. 3.7 illustrates the highly imbalanced class distribution of TreeAI-Swiss, with dominant species such as *Picea abies* and *Abies alba* contributing a large fraction of the samples, while several species appear only sparsely. This long-tailed distribution makes naive random splitting unsuitable, as it may lead to missing or underrepresented classes in validation and test sets.

We formulate dataset splitting as an integer linear programming (ILP) problem defined at the patch level. Each 60m $\times$ 60m patch is treated as an indivisible unit and is assigned to exactly one split (training, validation, or test). The desired split ratio be 70%, 15%, and 15%. For each patch, we first compute the frequency of all tree species contained within it. The optimization objective is to minimize the weighted L_1 deviation between the species distribution of each split and the overall dataset distribution, summed over all species and splits. To mitigate the dominance of frequent species, species-wise weights are introduced, where rarer species are assigned higher importance via an inverse-frequency weighting scheme.

The detailed implementation of the proposed ILP-based splitting algorithm is provided in the Appendix A.2.

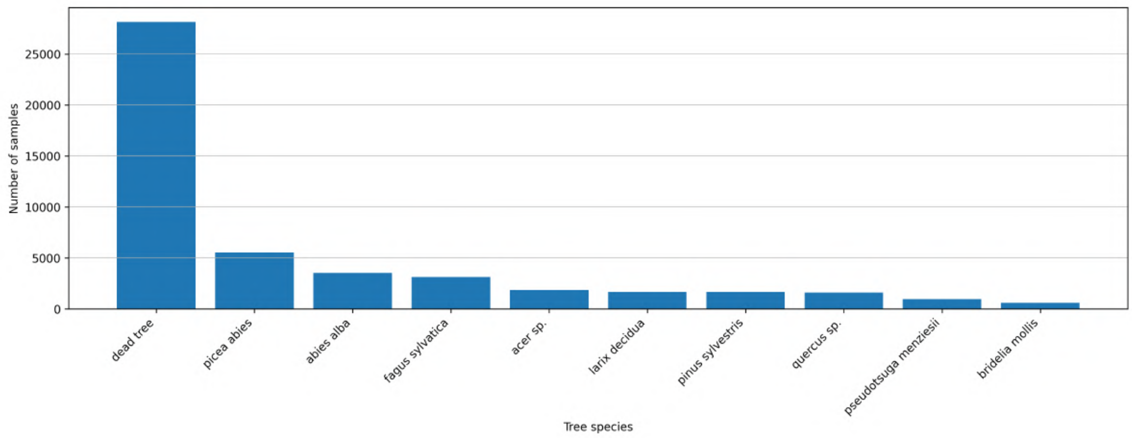


Figure 3.7: Tree species sample distribution in the TreeAI-Swiss dataset. The highly imbalanced and long-tailed distribution across species requires a balanced and spatially constrained dataset splitting strategy.

Chapter 4

Model and Experiments

To further examine whether multi-modal inputs can improve tree species classification accuracy and to analyze how different modalities interact, we not only construct a multi-modal, individual-tree-level dataset, namely TreeAI-Swiss, but also design a gated multi-modal fusion network. The proposed model is designed to be plug-in and plug-out, explicitly learning how different modalities contribute to tree species detection through learnable gating mechanisms.

4.1 Foundation Model Encoders

4.1.1 AnySat Visual Encoder

Although the proposed model is designed to fuse multiple modalities, visual-based inputs remain the most fundamental and informative sources for tree species classification. In particular, HR orthophoto mosaics and Sentinel-1/2 time series provide direct cues on canopy structure, which are strongly related to tree species characteristics.

As discussed in Section 2.3, recent remote sensing foundation models leverage large scale self-supervised pretraining (e.g., masked autoencoding) to learn generalizable representations from heterogeneous Earth observation data. In this work, we adopt AnySat (Astruc et al., 2025) as the visual encoder to extract feature representations from HR orthophoto imagery and Sentinel-1/2 time series.

AnySat follows a sensor-free design paradigm, meaning that it does not impose restrictions on the number or type of input modalities, as long as the modality has been observed during pretraining. We adopt the dense output mode of AnySat, which produces a feature map of size $1536 \times 30 \times 30$. This feature map is subsequently

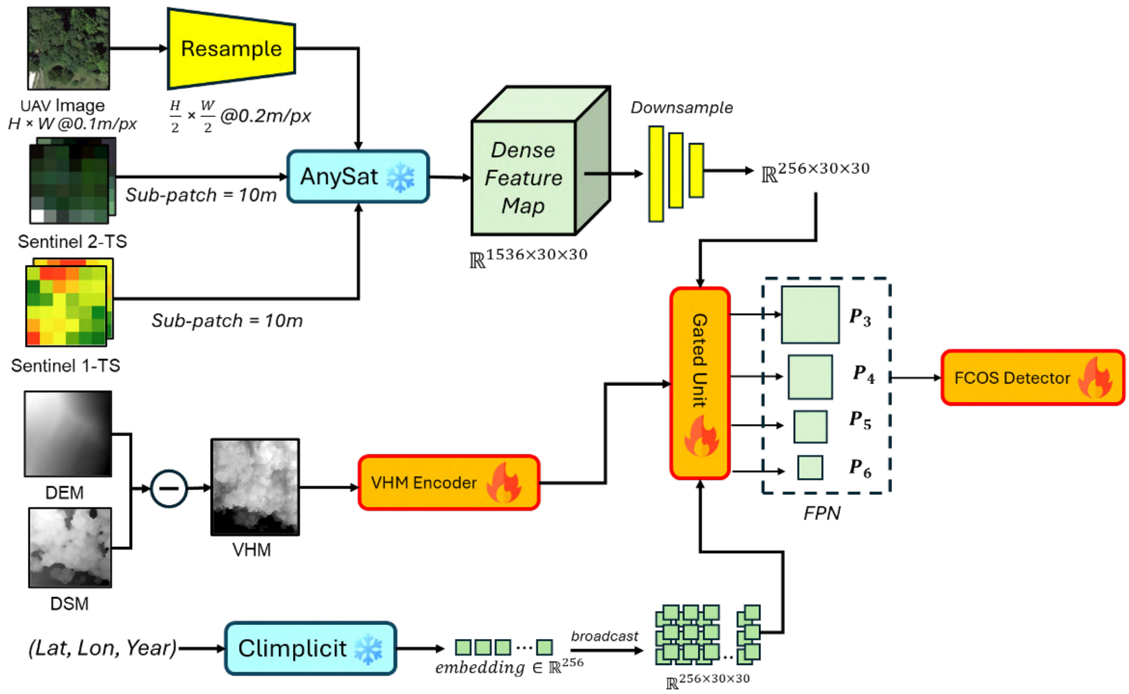


Figure 4.1: Model structure of gated multi-modal fusion network.

projected via a lightweight channel reduction layer to a $256 \times 30 \times 30$ representation, which serves as the visual input to the downstream multi-modal fusion module.

4.1.2 Climplicit Bioclim Encoder

Climplicit is a spatio-temporal geolocation encoder pretrained on CHELSA BIOCLIM to generate implicit climatic representations anywhere on Earth (Dollinger et al., 2025). Climplicit takes geographic coordinates and temporal information as input. We feed the latitude and longitude of each patch center and the corresponding observation year to obtain a 256-dimensional climatic embedding. Given the limited spatial extent of each 60m $\times$ 60m patch, we assume the climatic conditions within the patch is uniform. Therefore we broadcast the embedding to align with the 30×30 dense feature map produced by AnySat, resulting in a $256 \times 30 \times 30$ climate feature representation.

4.2 Vegetation Height Model Encoder

Inspired by Teng et al. (2025), we incorporate canopy height as an additional modality to improve tree crown detection and segmentation, especially in dense forests where crowns strongly overlap in RGB imagery. We compute a Vegetation Height

Model (VHM) by subtracting DEM from DSM, and yield height-above-ground estimates while reducing background terrain effects compared to using DSM or DEM alone. Following the same VHM encoder structure of Teng et al. (2025), we design a lightweight three-layer CNN to encode the VHM into a dense feature map. The encoder applies a 3×3 convolution with stride 2 to downsample the input, followed by a second 3×3 convolution and a final 1×1 projection to 256 channels, each with GELU activations and LayerNorm.

4.3 Gated Fusion Network

Although the proposed model integrates multiple modalities, the relative importance and interaction patterns among these modalities are still unknown. Different tree species may rely on different cues for discrimination.

Inspired by Mena et al. (2024), we design a gated multi-modal fusion module that adaptively learns the contribution of each modality.

Let $\{\mathbf{F}_i \in \mathbb{R}^{C \times H \times W}\}_{i=1}^{M}$ denote the modality-specific feature maps, where M is the number of modalities and all features are projected to a common channel dimension C. In our case, these modalities include visual features extracted by AnySat, vegetation height features encoded from the VHM, and climatic embeddings derived from Climplicit. For each modality, we apply global average pooling to obtain a compact descriptor:

$$\mathbf{z}_i = \frac{1}{HW} \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{F}_i(:, h, w), \tag{4.1}$$

where $\mathbf{z}_i \in \mathbb{R}^C$ summarizes the global response of modality i.

The pooled descriptors are concatenated and passed through a lightweight gating network to predict modality weights:

$$\boldsymbol{\alpha} = \mathrm{softmax}(\mathbf{W} \left[\mathbf{z}_1 \| \mathbf{z}_2 \| \dots \| \mathbf{z}_M\right]), \tag{4.2}$$

where $\boldsymbol{\alpha} \in \mathbb{R}^M$ and $\sum_{i=1}^{M} \alpha_i = 1$. Here, $\mathbf{W}$ denotes a learnable linear projection.

Finally, the fused feature map is computed as a weighted sum of those features:

$$\mathbf{F}_{\text{fused}} = \sum_{i=1}^{M} \alpha_i \, \mathbf{F}_i. \tag{4.3}$$

Table 4.1: Training configuration and hyperparameters used for the model training

Parameter	**Value**
Optimizer	Adam
Learning rate	1×10^{-3}
Batch size	16
Number of epochs	200
Number of data loader workers	4
Loss function	Focal loss
Sparse ignore threshold	0.3
Number of tree species classes	10
Fusion feature dimension	256
Sentinel-1 unit	dB

4.4 Training Details

TreeAI-Swiss is a sparsely annotated remote sensing dataset, where only a subset of tree crowns within each image patch is reliably labeled, while the remaining regions are unlabeled rather than true background. Treating all unlabeled locations as negative samples would introduce a large number of false negatives.

To address this issue, we introduce a sparse ignore threshold mechanism based on model prediction confidence. During training, candidate locations whose classification confidence falls within an uncertainty interval are excluded from loss computation. Instead of forcing such ambiguous locations to be classified as either foreground or background, we ignore them during optimization.

In addition, we adopt Focal Loss for classification to mitigate the severe foreground background imbalance inherent to dense, one–stage object detectors. Focal Loss down–weights well–classified easy negatives and concentrates learning on a sparse set of hard examples (Lin et al., 2017).

The model is trained end-to-end using the Adam optimizer with a fixed learning rate, and all training hyperparameters are summarized in Table 4.1.

Chapter 5

Results

5.1 ILP-based Splitting Algorithm

Table 5.1 shows the results of the proposed ILP-based dataset splitting algorithm. Although TreeAI-Swiss exhibits a long-tailed species distribution, the resulting train, validation, and test splits show highly consistent class proportions. This is evidenced by the stable test-to-train ratios reported in the last column, which remain nearly constant across all species.

A fair split helps ensure that the models, trained and tested on TreeAI-Swiss dataset, will be rated based on similar class distributions. Moreover, since the split is done at the patch-level and based on strict spatial exclusivity, the resulting dataset helps prevent the leakage of spatial information and also remains the spatial independence of the partitions.

TreeAI-Swiss consists of 7,292 spatially independent patches, each covering an area of $60\,\mathrm{m} \times 60\,\mathrm{m}$, resulting in a total spatial coverage of $26.25\,\mathrm{km}^2$ (equivalently 2,625.12 ha). The dataset includes 10 common Swiss tree species and contains more than 17,000 high-quality, expert-verified annotations under a sparse but reliable labeling scheme. Each patch integrates rich multi-modal remote sensing information, including high-resolution aerial orthophotos, topographic data (DEM/DSM), Sentinel-1 and Sentinel-2 time series, GSE, and auxiliary bioclimatic variables. TreeAI-Swiss also supports a wide range of downstream forest-related tasks, such as tree species classification, crown detection and segmentation, canopy height modeling, and forest health monitoring.

Table 5.1: Species-wise sample distribution across dataset splits in TreeAI-Swiss. Although the overall distribution is highly imbalanced, the relative split ratios remain consistent across species.

Species	**Train**	**Validation**	**Test**	**Test / Train**
Dead tree	9854	2112	2111	0.214
Picea abies	1940	416	415	0.214
Abies alba	1237	265	265	0.214
Fagus sylvatica	1094	234	235	0.215
Quercus sp.	555	119	119	0.214
Acer sp.	646	138	139	0.215
Larix decidua	579	124	124	0.214
Pinus sylvestris	574	123	123	0.214
Pseudotsuga menziesii	334	71	72	0.216
Bridelia mollis	206	44	44	0.214

5.2 Evaluation Metrics

5.2.1 Hit Recall

Following prior work on individual tree detection and crown delineation (Safonova et al. (2021b); Reiersen et al. (2022); Troles et al. (2024)), we adopt *Hit Recall (HR)* as a primary metric to evaluate tree-level detection performance.

In these studies, a reference tree crown is considered correctly detected if at least one predicted instance sufficiently overlaps with the ground-truth crown. This definition emphasizes recall at the individual-tree level and is particularly suitable for dense forest scenes and sparse annotation settings.

Formally, let G denote the set of ground-truth tree crowns and P the set of predicted instances. A ground-truth crown $g \in G$ is counted as a hit if

$$\max_{p \in P} \mathrm{IoU}(g, p) \geq \tau, \tag{5.1}$$

where τ is a predefined overlap threshold. Here we use 0.4 as the threshold. The Hit Rate is then defined as

$$\mathrm{HR}(\tau) = \frac{1}{|G|} \sum_{g \in G} \mathbb{I}\left[\max_{p \in P} \mathrm{IoU}(g, p) \geq \tau\right], \tag{5.2}$$

where $\mathbb{I}[\cdot]$ is the indicator function.

Method Setup	Input Modality				Temporal Scale	Detection	Species–Genus (10 cls)	
	RGB	S1/2	VHM	Clim.		HR@0.4	Acc.	F1
RGB Baseline	✓				Month	0.652	0.566	0.565
+ S1/2	✓	✓			Month	0.672	0.572	0.607
+ S1/2 + VHM	✓	✓	✓		Month	0.636	0.593	0.609
+ S1/2 + Climate	✓	✓		✓	Month	**0.704**	0.639	0.656
Impact of Temporal Duration (with all modalities)								
Full Model (Annual)	✓	✓	✓	✓	Year	0.653	0.562	0.564
Full Model (Quarterly)	✓	✓	✓	✓	Quarter	0.659	0.559	0.581
Full Model (Monthly)	✓	✓	✓	✓	Month	0.660	**0.666**	**0.677**

Table 5.3: Detection and species–genus classification performance under different modality configurations and temporal scales.

Method Setup	Input Modality				Temporal Scale	Detection	Family (5 classes)	
	RGB	S1/2	VHM	Clim.		HR@0.4	Acc.	F1
RGB Baseline	✓				Month	0.652	0.738	0.758
+ Sentinel-1/2	✓	✓			Month	0.672	0.674	0.717
+ Sentinel-1/2 + VHM	✓	✓	✓		Month	0.636	0.751	0.764
+ Sentinel-1/2 + Climate	✓	✓		✓	Month	**0.704**	0.743	0.767
Impact of Temporal Resolution (All Modalities Enabled)								
Full Model (Annual)	✓	✓	✓	✓	Year	0.653	0.631	0.659
Full Model (Quarterly)	✓	✓	✓	✓	Quarter	0.659	0.722	0.744
Full Model (Monthly)	✓	✓	✓	✓	Month	0.660	**0.773**	**0.793**

Table 5.4: Detection and family-level classification performance under different modality combinations and temporal resolutions.

5.2.2 Sparse Accuracy and F1–Score

Due to the sparse annotation nature of the TreeAI-Swiss dataset, classification performance is evaluated only on ground-truth tree crowns that are successfully detected. Specifically, a predicted instance contributes to the classification metrics only if it is matched to a ground-truth crown with an IoU larger than a predefined threshold.

Sparse accuracy is computed as the classification accuracy over the set of matched ground-truth instances. To account for the severe class imbalance and long-tailed species distribution in TreeAI-Swiss, we further report the *macro* F1-score, where F1-scores are first computed independently for each tree species and then averaged across classes with equal weight.

Figure 5.1: Confusion matrix for the best-performing configuration (monthly temporal aggregation with all modalities enabled).

5.3 Tree Species Detection

All experiments are formulated as an object detection task and implemented using an FCOS-style detection head (Tian et al., 2019). The model is trained end-to-end using the training protocol, optimization settings, and loss configuration described in Section 4.4. Detection performance is evaluated using HR at an IoU threshold of 0.4 (HR@0.4), while classification performance is assessed using sparse Accuracy and Macro F1-score to account for the long-tailed class distribution of the TreeAI-Swiss dataset. Although training is consistently performed with ten species–genus level classes, evaluation is additionally conducted at the family level by mapping species-level predictions to their corresponding five taxonomic families.

5.3.1 Species-Genus Level

At the species–genus level, adding Sentinel-1/2 data to the RGB baseline increases HR@0.4 from 0.652 to 0.672 and improves sparse accuracy from 0.566 to 0.572, with a more noticeable gain in F1-score (0.565 to 0.607). However, incorporating the vegetation height model (VHM) does not yield further improvements in detection performance. Specifically, HR@0.4 decreases from 0.672 to 0.636 when VHM is added, while accuracy (0.593) and F1–score (0.609) show only marginal changes compared to the RGB+S1/2 setting.

In contrast, adding climatic embeddings leads to the strongest detection performance at this level, increasing HR@0.4 to 0.704. Classification metrics also improve, with accuracy rising to 0.639 and F1-score to 0.656. These results indicate that, at the species–genus level, different modalities contribute unevenly to detection and classification, with height information showing limited benefit under the current configuration.

5.3.2 Family Level

At the family level, improvements from additional modalities are more consistent. Compared to the RGB baseline, adding Sentinel-1/2 increases accuracy from 0.738 to 0.674 and F1-score from 0.758 to 0.717, followed by a further increase when VHM is included (accuracy 0.751, F1-score 0.764). Unlike the species–genus setting, the inclusion of VHM leads to clear gains in family-level classification.

The best family-level results are obtained when climatic embeddings are added, achieving an accuracy of 0.743 and an F1-score of 0.767, while maintaining the highest detection recall (HR@0.4 = 0.704).

5.3.3 Temporal Resolution

The impact of temporal resolution is evaluated by comparing annual, quarterly, and monthly temporal aggregations when all modalities are enabled. At the species–genus level, monthly inputs achieve the highest classification performance, with sparse Accuracy improving from 0.562 (annual) and 0.559 (quarterly) to 0.666, and Macro F1-score increasing from 0.564 and 0.581 to 0.677, while HR@0.4 shows only marginal variation (from 0.653 to 0.660). A similar trend is observed at the family level, where monthly aggregation yields the strongest results, reaching an accuracy of 0.773 and a Macro F1–score of 0.793, compared to 0.631/0.659 for annual and 0.722/0.744 for quarterly inputs. These results indicate that finer temporal resolution primarily benefits classification performance, whereas detection performance remains comparatively stable across different temporal scales.

5.4 Discussion

This work investigates individual tree detection through a multi-modal learning framework built upon the TreeAI-Swiss dataset.

The addition of further modalities, apart from high-resolution RGB imagery, improves detection performance, though the degree of improvement depends on the task and level of taxonomic identification. The addition of time series from Sentinel-1/2 and climatic embeddings shows a remarkable improvement, especially in terms of recall for detection as well as robustness for classification. The outcome suggests that there is complementarity in satellite-based data, including spectra, time, and climatic information, apart from RGB-based appearance features, since species distribution in ecology has been known to rely on canopy texture, in addition to other features, such as phenology, structure, and climatic constraints.

While the addition of information about vegetation height has been found useful in improving crown detection in a dense forest scenario, its role in the current set of experiments seems to be more task and scale dependent. Within the species/genus category, adding the VHM does not necessarily lead to improved detection or classification performance. This might be because the VHM encoder used in the current experiment still has some limitations in capturing the finer details of height information, which are of more relevance in distinguishing between species. However, the improvement in the family category suggests otherwise.

On classification tasks, in all experiments conducted, the classification performance with monthly temporal inputs outperforms all other aggregation periods, including annual and quarterly, with little change in the performance for detection tasks. This shows that higher temporal resolution is very beneficial for phenological pattern recognition to facilitate species or family classification.

Chapter 6

Conclusions and Limitations

6.1 Conclusions

In this research, we propose TreeAI-Swiss, a high-quality multi-modal dataset specifically designed for individual-tree assessment in complex forested settings. The dataset incorporates multi-modal remote sensing data such as high-resolution aerial orthoimages, topographic data (DEM/DSM-based vegetation height models), time-series Sentinel-1 and Sentinel-2 images, Google satellite embeddings, and auxiliary bioclimatic variables. A semi-automatic annotation framework was established using field measurements and crown delineation based on SAM3 to facilitate the creation of accurate instance-level annotations. To ensure independence of instances and balance of classes with a long-tailed species distribution, we additionally introduce an ILP-based splitting approach for our dataset.

On top of such a dataset, we introduce the gated multi-modal fusion model leveraging a frozen AnySat visual encoder, light-weight VHM encoders, as well as an adaptive gating module to represent the roles of different modalities. The model enables adaptive re-weighting of complementary sources of spectral, temporal, structural, and environmental information. The incorporation of vegetation height information and embeddings of climatic factors are added as other sources of modalities.

Experiments have shown that multi-modal information improves the performance of detection and classification. For species-genus classification, it shows that the contribution of different modalities varies in terms of its performance for detection recall and fine-classification tasks. Although the vegetation height model almost does not improve performance at the species classification task, it makes a better contribution at the family-level classification task. In terms of the temporal aggregation, it shows that the use of the monthly series improves performance compared to the quarterly or annual series, when all the modalities are enabled.

Also, per-species assessment (Tables A.3 and A.4) shows a diverse performance trend for different tree species. In particular, it is evident that the dead trees class has high accuracy and F1-score for all scenarios, almost reaching perfection for the fully monthly approach. This kind of performance demonstrates the good separability between dead trees in the designed multi-modal feature space, which is presumably caused by the difference of spectral and spatial properties of dead trees. The reliability for dead tree detection confirms the applicability not only for mapping but also for monitoring forest condition assessment tasks.

In conclusion, the TreeAI-Swiss system offers a complete reference framework for examining multi-modal and instance-level forest analysis, and the proposed approach presents a flexible reference platform for the study of adaptive fusion strategies in complex forest environments.

6.2 Limitations

There exist some shortcomings in the presented research that should be pointed out. First, although the quality of annotations is very high, the dataset could also be enriched with full annotations.

Second, although Swisstopo-based high-resolution data sources are well coregistered, Sentinel-based data sources are only reprojected in a common coordinate reference system without strict sub-pixel coregistration; while small misalignments are less probable to influence learning in the patch scale used in this study, stricter cross-sensor alignment may also help in improving model robustness.

Thirdly, the proposed approach is a fully supervised system and there is no specific handling of strong class imbalance aside from the evaluation metrics. It may also be beneficial to incorporate semi-supervised or weakly supervised learning techniques in order to derive further benefits from the unlabeled or partially labeled samples, particularly in the case of rare species.

Concluding, the TreeAI-Swiss dataset is now limited to the country of Switzerland. It should be noted that the results achieved within the framework of this study are specific to Swiss forests. Further analysis related to other regions around the world would, undoubtedly, turn out to be productive as well.

Appendix A

Supplementary Materials

A.1 Summary of aerial image-based forest datasets

A.2 Implementation of ILP-based Splitting Algorithm

Let $x_{g,k} \in \{0, 1\}$ indicate whether patch g is assigned to split $k \in \{\text{train}, \text{val}, \text{test}\}$. Let $C_{g,s}$ denote the number of trees of species s contained in patch g. The achieved species count in split k is then

$$S_{k,s} = \sum_{g} x_{g,k} \, C_{g,s},$$

while the target count is defined as

$$T_{k,s}^{\text{target}} = r_k \, T_s,$$

where r_k denotes the desired split ratio and $T_s = \sum_g C_{g,s}$ is the global count of species s. To ensure that each split closely matches the overall species distribution, we minimize the weighted absolute deviation between achieved and target counts:

$$\min_{\{x_{g,k}\}} \sum_{k} \sum_{s} w_s \left| S_{k,s} - T_{k,s}^{\text{target}} \right|, \tag{A.1}$$

where w_s assigns higher importance to rare species.

Table A.1: Summary of aerial image-based forest datasets for tree-related tasks.

Dataset	**Area**	**GSD / Point Density**	**# Samples / Patches**	**Task Type**
TreeSatAI	Central Europe (Germany)	20 cm/pix (aerial), 10 m (S1, S2)	50381	Tree Species Classification
Perm_Krai	Russia	7 cm/pix, 37 pts/m^2 (LiDAR)	3600	Individual Tree Detection
PureForest	France	20 cm/pix, 40 pts/m^2 (LiDAR)	135,569	Tree Species Classification
NEON Tree	United States	0.1 m/pix (aerial), 4–6 pts/m^2 (LiDAR)	30975	Individual Tree Detection
ReforesTree	Ecuador	2 cm/pix	4600	Individual Tree Detection
Woody	Chile	3–4 cm/pix	3249	Semantic Segmentation
Spekboom	South Africa	approx. 1 cm/pix	32 (orthomosaics)	Semantic Segmentation
Waititu	New Zealand	2–3 cm/pix	5000	Semantic Segmentation
OAM-TCD	Global Urban Areas	10 cm/pix	5072	Instance Segmentation
RT-Trees	Central Alberta, Canada	3 cm/pix	49,879	Individual Tree Detection
BAMFORESTS	Bamberg, Germany	1–2 cm/pix	27160	Instance Segmentation
OliveTreeCrownsDb	Meknes region, Morocco	1.78 cm/pix	33,448	Individual Tree Detection
SELVABOX	Panama, Brazil, Ecuador	1.2–5.1 cm/pix	83,137	Individual Tree Detection
BCI dataset	Panama	4 cm/pix	approx. 27,000	Instance Segmentation
Quebec Trees Dataset	Quebec, Canada	1.9 cm/pix	22,933	Instance Segmentation
Quebec Plantations dataset	Quebec, Canada	5 mm/pix	Not mentioned	Instance Segmentation

A.3 Per-species Classification Results

Table A.3: Per-species F1-scores under the best-performing setting (Full model with all modalities and monthly Sentinel inputs. Best results are highlighted in **bold**.

Species	*P. abies*	*P. sylv.*	*L. dec.*	*F. sylv.*	*Dead tree*	*A. alba*	*P. menz.*	*Acer sp.*	*Quercus*	*B. mollis*
Class Dist. (%)	6.87	2.86	3.74	5.99	66.59	5.06	2.11	2.95	2.46	1.36
RGB	0.595	**0.559**	0.804	0.683	**0.997**	0.577	0.286	0.460	0.000	0.692
RGB + S1/2	0.489	0.489	0.713	0.667	0.988	0.598	0.737	0.495	0.383	0.511
RGB + S1/2 + VHM	0.573	0.395	0.727	0.576	0.990	**0.620**	0.456	**0.524**	0.409	0.732
RGB + S1/2 + Clim.	**0.624**	0.463	**0.847**	0.604	0.992	0.509	0.769	0.508	0.509	0.758
Full (Annual)	0.539	0.430	0.637	0.612	0.993	0.574	0.630	0.176	0.493	0.537
Full (Quarterly)	0.584	0.388	0.312	0.690	0.990	0.583	0.651	0.427	0.553	0.632
Full (Monthly)	0.610	0.459	0.697	**0.697**	0.995	0.617	**0.821**	0.466	**0.576**	**0.833**

Table A.4: Per-species classification accuracy under the best-performing setting (Full model with all modalities and monthly Sentinel inputs. Best results are highlighted in **bold**.

Species	*P. abies*	*P. sylv.*	*L. dec.*	*F. sylv.*	*Dead tree*	*A. alba*	*P. menz.*	*Acer sp.*	*Quercus*	*B. mollis*
Class Dist. (%)	6.87	2.86	3.74	5.99	66.59	5.06	2.11	2.95	2.46	1.36
RGB	0.690	**0.702**	0.796	0.782	**0.999**	0.457	0.184	**0.465**	0.000	0.581
RGB + S1/2	0.519	0.355	0.610	0.656	0.997	**0.791**	0.636	0.397	0.367	0.387
RGB + S1/2 + VHM	**0.729**	0.306	0.622	0.519	0.994	0.691	0.456	0.458	0.409	0.750
RGB + S1/2 + Clim.	0.671	0.373	**0.822**	0.612	**0.999**	0.531	0.769	0.456	0.465	0.694
Full (Annual)	0.526	0.318	0.653	0.548	0.995	0.781	0.630	0.118	0.661	0.393
Full (Quarterly)	0.642	0.338	0.250	**0.812**	0.990	0.661	0.562	0.344	0.448	0.545
Full (Monthly)	0.686	0.339	0.635	0.669	0.999	0.678	**0.812**	0.358	**0.679**	**0.807**

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